TA Session I: Theory Handout

Microeconomics I, Part B — Choice under Uncertainty

The concepts behind Exercises 8, 9, 4, and 10

This handout collects the results you need for the session's exercises. Each section ends with a pointer to where the idea is used. The suggested order of work is 8, 4, 9, 10: expected utility first, then measures of risk aversion, then comparisons across agents and levels of wealth.

1 Lotteries and expected utility

Lotteries. With a finite set of outcomes $\{x_1, \dots, x_N\}$, a *simple lottery* is a probability vector $L = (p_1, \dots, p_N)$, with $p_n \geq 0$ and $\sum_n p_n = 1$. A *compound lottery* gives simple lotteries $L_1, \dots, L_K$ with probabilities $\alpha_1, \dots, \alpha_K$. Its *reduced lottery* is $\sum_k \alpha_k L_k$: what matters for choice is only the final probability of each outcome.

The von Neumann–Morgenstern axioms. A preference relation $\succsim$ over lotteries satisfies

- *rationality*: it is complete and transitive;
- *continuity*: if $L \succsim L' \succsim L''$, then $L' \sim \alpha L + (1-\alpha)L''$ for some $\alpha \in [0,1]$;
- *independence*: $L \succsim L'$ if and only if $\alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''$ for every L'' and every $\alpha \in (0,1]$.

Expected utility theorem. Under these axioms there are numbers $u_1, \dots, u_N$ (one per outcome) such that

$$L \succsim L' \iff \sum_n p_n u_n \;\geq\; \sum_n p'_n u_n.$$

The representation $U(L) = \sum_n p_n u_n$ is unique up to a *positive affine transformation*: $\tilde{u}_n = \beta u_n + \gamma$ with $\beta > 0$ represents the same preferences.

Two consequences are used constantly.

(i) **Linearity in probabilities.** $U(\alpha L + (1-\alpha)L') = \alpha U(L) + (1-\alpha)U(L')$. A lottery over lotteries is evaluated by the probability-weighted average of their utilities.

(ii) **Free normalization.** Because of affine uniqueness, you may set the utility of the worst outcome to 0 and of the best to 1. The remaining utilities are then pinned down by indifference conditions.

Used in: Exercise 8. Indifference between Schools 1 and 3 gives one linear equation in (u_A, u_B, u_C); the normalization $u_A = 0$, $u_B = 1$ fixes the rest. School 4 and the admission lottery are then evaluated by linearity.

2 Monetary lotteries: computing expected utility

When outcomes are amounts of money, the lottery is a random variable X and preferences are represented by $U(X) = \mathbb{E}[u(X)]$. The function u is called the *Bernoulli utility function*, after Daniel Bernoulli. The name refers to the utility of each sure outcome, *not* to the Bernoulli probability distribution.

How the expectation is computed depends only on the distribution of X:

$$\mathbb{E}[u(X)] = \sum_n p_n\, u(x_n) \qquad\qquad \mathbb{E}[u(X)] = \int u(x)\, f(x)\, dx$$

discrete X — continuous X with density f

The integral is the limit of the sum: $f(x)\,dx$ plays the role of the probability of a small interval around x. For X uniform on $[\underline{x}, \bar{x}]$ the density is constant, $f(x) = 1/(\bar{x} - \underline{x})$; on $[0,1]$ it equals 1.

Used in: Exercise 9. $X \sim U[0,1]$ requires an integral, $\mathbb{E}[X^a] = \int_0^1 x^a\, dx = 1/(a+1)$. Y is discrete (in fact a Bernoulli(2/3) random variable), so a sum suffices.

3 Risk aversion

Definition. The agent is *risk averse* if, for every lottery X, receiving $\mathbb{E}X$ for sure is at least as good as X:

$$u(\mathbb{E}X) \;\geq\; \mathbb{E}[u(X)].$$

By Jensen's inequality this holds for every lottery if and only if u is *concave*. Strict concavity gives strict risk aversion, linear u gives risk neutrality, and convex u gives risk loving.

Certainty equivalent and risk premium. The *certainty equivalent* $c(X)$ is the sure amount giving the same utility as the lottery, and the *(monetary) risk premium* is the expected payoff the agent would give up to remove the risk:

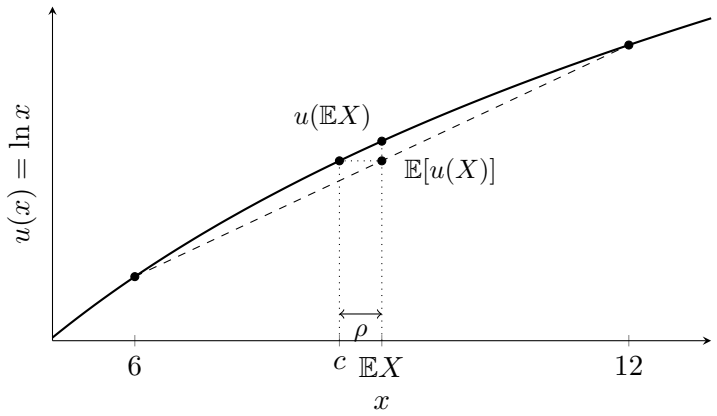
$$u\big(c(X)\big) = \mathbb{E}[u(X)], \qquad \text{i.e. } c(X) = u^{-1}\big(\mathbb{E}[u(X)]\big); \qquad\qquad \rho(X) = \mathbb{E}X - c(X).$$

A risk-averse agent has $c(X) \leq \mathbb{E}X$, that is $\rho(X) \geq 0$.

Probability premium. Fix wealth x and a symmetric gamble $\pm\varepsilon$. The *probability premium* $\pi(x, \varepsilon, u)$ is the extra probability on the good outcome that makes the agent indifferent to keeping x for sure:

$$u(x) = \left(\tfrac{1}{2} + \pi\right) u(x+\varepsilon) + \left(\tfrac{1}{2} - \pi\right) u(x-\varepsilon).$$

Again $\pi \geq 0$ under risk aversion. Note that ρ is an amount of money and π is a probability, so their sizes cannot be compared with each other directly.



Lottery paying 6 or 12 with equal probability under $u = \ln x$. Expected utility lies on the chord, below the curve, so $c = \sqrt{72} \approx 8.49 < 9 = \mathbb{E}X$.

Used in: Exercise 4(a)–(b) (certainty equivalent and $\rho_A = 9 - c_A$), Exercise 4(d) (probability premium), Exercise 10(a)–(b) (certainty equivalent with $\sqrt{x}$).

4 Measuring risk aversion: Arrow–Pratt

Concavity says *whether* an agent is risk averse; the Arrow–Pratt coefficients say *how much*. For twice-differentiable increasing u,

$$r_A(x) = -\frac{u''(x)}{u'(x)} \quad \text{(absolute)}, \qquad r_R(x) = x\, r_A(x) = -\frac{x\, u''(x)}{u'(x)} \quad \text{(relative)}.$$

Dividing by u' makes r_A invariant to positive affine transformations of u, as it must be, since those do not change preferences.

Small risks. For a small risk Z with $\mathbb{E}Z = 0$ added to wealth x, a second-order Taylor expansion gives

$$\rho \approx \tfrac{1}{2}\,\mathrm{Var}(Z)\, r_A(x), \qquad \pi(x, \varepsilon, u) \approx \tfrac{1}{4}\,\varepsilon\, r_A(x).$$

So $r_A(x)$ is (twice) the premium per unit of variance at wealth x.

Check with Exercise 4. At $x = 9$, $r_A(9) = 1/9$. For asset A, $\mathrm{Var} = 6$, so $\rho \approx \frac{1}{2} \cdot 6 \cdot \frac{1}{9} \approx 0.333$ (exact: 0.3465). For the ± 3 gamble, $\pi \approx \frac{1}{4} \cdot 3 \cdot \frac{1}{9} \approx 0.0833$ (exact: 0.0850).

Comparing agents (Pratt's theorem). For two utilities u_1, u_2, the following are equivalent:

(i) $r_A^1(x) \geq r_A^2(x)$ for every x;

(ii) $u_1 = \varphi(u_2)$ for some increasing concave φ ("u_1 is more concave");

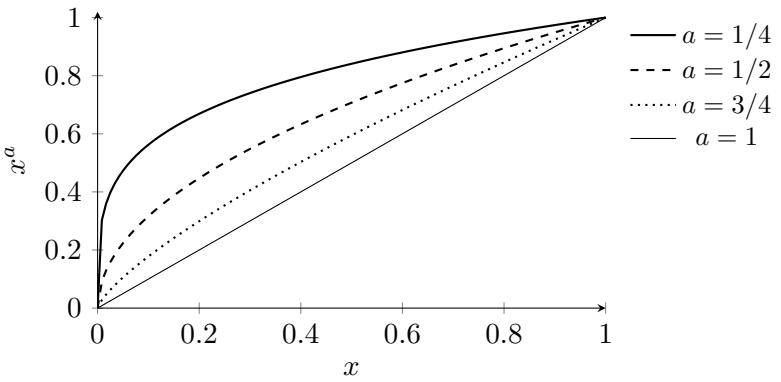
(iii) $c_1(X) \leq c_2(X)$, equivalently $\rho_1(X) \geq \rho_2(X)$, for every lottery X;

(iv) $\pi_1(x, \varepsilon) \geq \pi_2(x, \varepsilon)$ for every x and ε.

All of these mean that agent 1 is *more risk averse* than agent 2. Statement (iii) compares a lottery with a sure amount. It does *not* say how the two agents rank two different risky lotteries, which in general requires the whole distributions, not just means and variances.

Common functional forms.

$u(x)$	$r_A(x)$	$r_R(x)$	Class	Exercise
x^a, $0 < a < 1$	$(1-a)/x$	$1-a$	CRRA, DARA	9
$\sqrt{x}$	$1/(2x)$	$1/2$	CRRA, DARA	10
$\ln x$	$1/x$	1	CRRA, DARA	4
$-e^{-\theta x}$, $\theta > 0$	θ	θx	CARA	—
x	0	0	risk neutral	—



Smaller a means a more concave u and a larger $r_A = (1-a)/x$: a more risk-averse agent.

Used in: Exercise 9. For $a < b$, $x^a = (x^b)^{a/b}$ and $t \mapsto t^{a/b}$ is concave, so by Pratt's theorem a smaller a means more risk aversion. In that exercise, $\mathbb{E}[u(Y)] = 2/3$ does not depend on a (since $0^a = 0$, $1^a = 1$), while $\mathbb{E}[u(X)] = 1/(1+a)$ rises as a falls. The more risk-averse agents ($a < 1/2$) prefer the spread-out X; the less risk-averse ones prefer the higher-mean Y.

5 Risk and wealth

Background wealth. If the agent already holds wealth w, a lottery X is evaluated as an *addition* to wealth, $\mathbb{E}[u(w+X)]$. Its value to the agent then depends on w.

DARA, CARA, IARA. The sign of $r_A'(x)$ says how the cost of a *given additive* risk changes with wealth:

- *decreasing* absolute risk aversion (DARA): richer agents demand a smaller premium for the same risk ($\ln x$, x^a, $\sqrt{x}$);
- *constant* (CARA): the premium does not depend on wealth ($-e^{-\theta x}$);
- *increasing* (IARA): richer agents demand a larger premium (usually considered implausible).

Similarly, constant relative risk aversion (CRRA) means the agent's attitude to risks *proportional* to wealth does not change with wealth. For CRRA utility, scaling a lottery scales its certainty equivalent: $c(hX) = h\,c(X)$ for $h > 0$.

Minimum selling price. An agent with wealth w who owns X accepts to sell it at any price $p \geq p_{\min}$, where

$$u(w + p_{\min}) = \mathbb{E}[u(w+X)] \quad \Longleftrightarrow \quad p_{\min} = u^{-1}\big(\mathbb{E}[u(w+X)]\big) - w.$$

In general $p_{\min} \neq c(X) = u^{-1}(\mathbb{E}[u(X)])$: they coincide for every w only under CARA (or risk neutrality). Under DARA, extra wealth makes the risk cheaper to bear, so $p_{\min}$ rises with w towards $\mathbb{E}X$.

Used in: Exercise 4(c), where $c(h_A X_A) = h_A\,c_A$ because $\ln x$ is CRRA, so $h_A = 9/c_A$. Exercise 10(d)–(e): with $\sqrt{x}$ and $w = 12$, $c = 4 < p_{\min} \approx 4.85 < \mathbb{E}X = 5$, and the risk premium falls from 1 to 0.15. The small-risk formula shows why: $\operatorname{Var} X = 10$, so $\rho \approx \frac{1}{2} \cdot 10 \cdot r_A(17) = \frac{10}{68} \approx 0.147$ at $w = 12$. Without background wealth the approximation is poor (0.5 against the exact 1), because the risk is large relative to the payoff.

6 Summary: concepts by exercise

Exercise	Concepts
8	Expected utility theorem; normalization by affine transformations; linearity in probabilities; reduction of compound lotteries.
4	Certainty equivalent; monetary risk premium; CRRA scaling of a lottery; probability premium; r_A and r_R for $\ln x$.
9	Expected utility of a continuous lottery; comparing two risky lotteries; curvature as risk aversion; r_A for x^a.
10	Definition of the certainty equivalent; r_A for $\sqrt{x}$; background wealth and selling price; DARA.